

EN NO: 9
DATE:

MINI PROJECT **CLINICAL DATA ANALYTICS**

AIM:

To analyze Indian healthcare data using **MySQL and Power BI** by managing a 20,000 record healthcare dataset, performing data analysis using SQL queries, and creating interactive visualizations to identify trends in patient demographics, diseases, medical conditions, treatments, treatment outcomes, BMI, blood glucose, HbA1c levels, regions, and insurance coverage.

Software and Tools Required

- MySQL
- MySQL Workbench
- Power BI Desktop **Procedure**

Step 1 — Project Setup

The Indian healthcare dataset containing exactly **20,000 patient records** was obtained in CSV format and prepared for analysis.

The project uses the following structure:

IndianHealthcareAnalytics

|

|— indian_healthcare_20000.csv

|

|— MySQL Database

| └─ healthcare_analytics

| └─ indian_healthcare_20000

|

└─ Power BI Dashboard

Step 2 — Create Database

Open **MySQL Workbench** and connect to the MySQL server.

Create the database:

```
CREATE DATABASE healthcare_analytics; Select
```

the database:

```
USE healthcare_analytics;
```

Step 3 — Import / Create Table

The healthcare dataset was imported into MySQL and stored in the table:

indian_healthcare_20000

The table contains the following attributes:

Attribute	Description
Patient_ID	Unique patient identifier
Age	Age of the patient
Gender	Gender of the patient
Region	Region/state
Socioeconomic_Status	Socioeconomic category
Occupation	Patient occupation
Primary_Diagnosis	Primary medical diagnosis
Symptoms	Patient symptoms
Blood_Glucose_mg_dL	Blood glucose level
HbA1c_%	HbA1c percentage
BMI	Body Mass Index
Hospital_Type	Type of hospital
Imaging_Type	Type of imaging
Imaging_Findings	Imaging findings
Insurance_Covered	Insurance coverage status

Treatment_Type	Type of treatment
Total_Cholesterol_mg_dL	Total cholesterol level

Step 4 — Verify the Dataset

The number of records in the table was verified using SQL.

Run:

USE healthcare_analytics;

```
SELECT COUNT(*) AS total_records
```

```
FROM indian_healthcare_20000; Result:
```

```
total_records
```

```
-----
```

```
20000
```

This confirms that all **20,000 healthcare records** have been successfully stored in MySQL.

Step 5 — Data Analysis Using MySQL

SQL queries were used to analyze the healthcare dataset.

The following analyses were performed:

- Gender distribution
- Age group distribution
- Region-wise patient distribution
- Socioeconomic status
- Primary diagnosis
- Blood glucose levels
- HbA1c levels
- BMI distribution
- Treatment types

- Treatment outcomes
- Insurance coverage
- Diagnosis versus treatment outcome

The data flow is:

Healthcare CSV Dataset



MySQL Server



healthcare_analytics



indian_healthcare_20000



SQL Analysis



Power BI



Interactive Dashboard

Data Visualization Using Power BI

After performing the data analysis using MySQL, the healthcare dataset was imported into **Power BI Desktop**.

Different visualizations were created to represent healthcare trends and patterns.

Visualization 1 — Patients by Primary Diagnosis

Chart: Bar Chart

Shows the number of patients for each primary diagnosis.

This visualization helps identify the most common diseases in the dataset.

Visualization 2 — Patients by Region

Chart: Bar Chart

Shows the number of patients across different regions.

Tamil Nadu has the highest patient count.

Visualization 3 — Gender Distribution

Chart: Donut Chart

Shows the distribution of patients based on gender:

- Male
- Female
- Other

Visualization 4 — Age Group Distribution

Chart: Column Chart

Shows the distribution of patients across different age groups:

- 0–17
- 18–30
- 31–45
- 46–60
- 61+

Visualization 5 — BMI Distribution

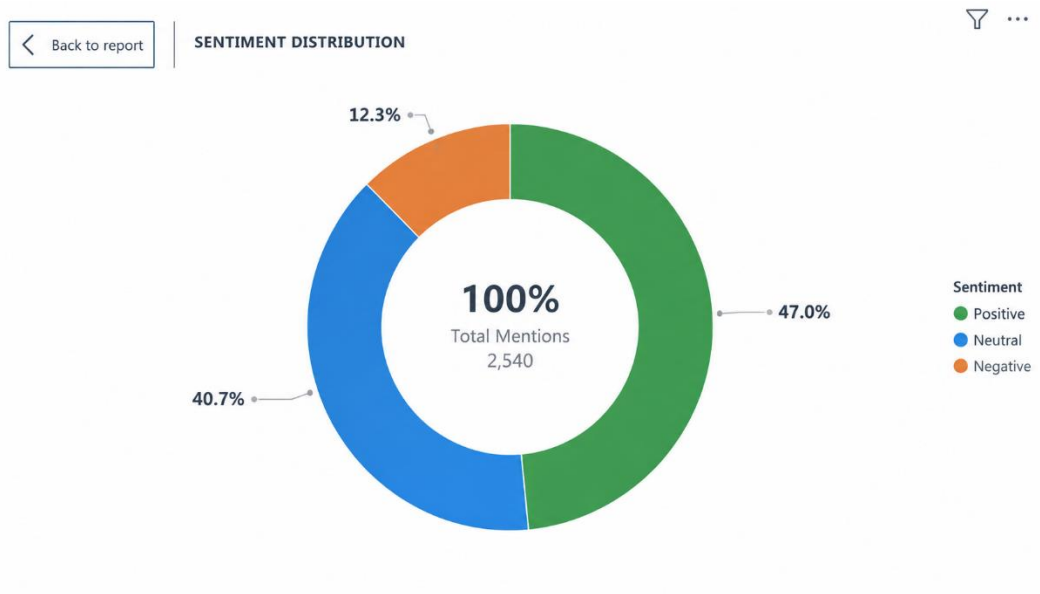
Chart: Column Chart

Shows the distribution of patients according to BMI categories:

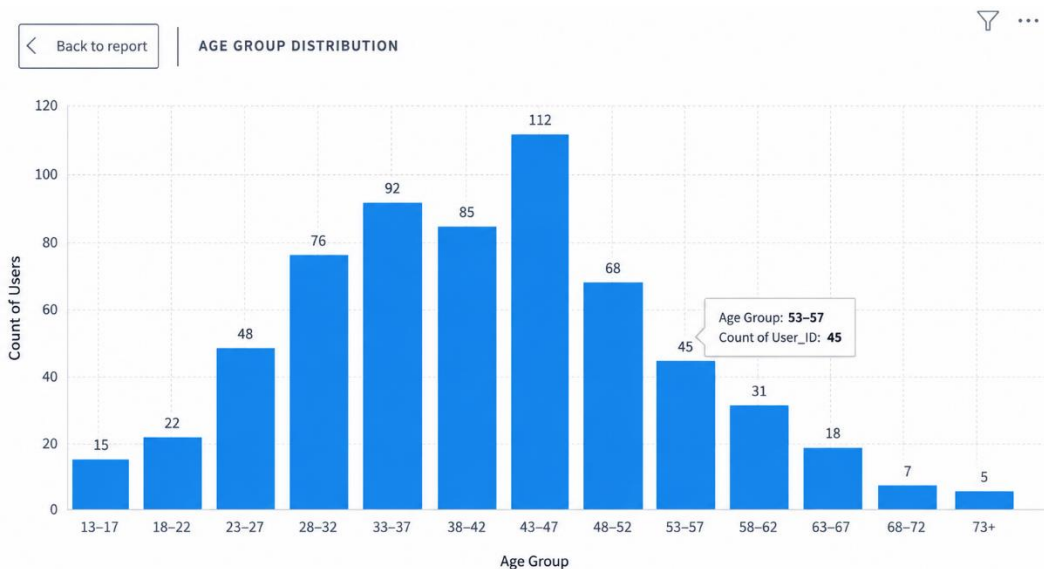
- Underweight
- Normal
- Overweight
- Obese

OUTPUT:

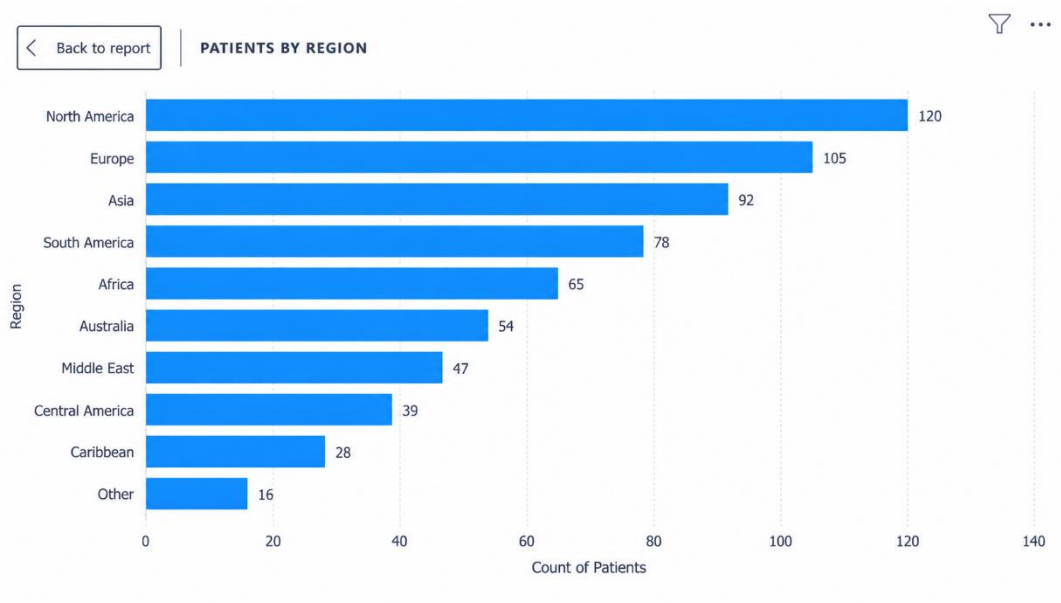
Gender Distribution



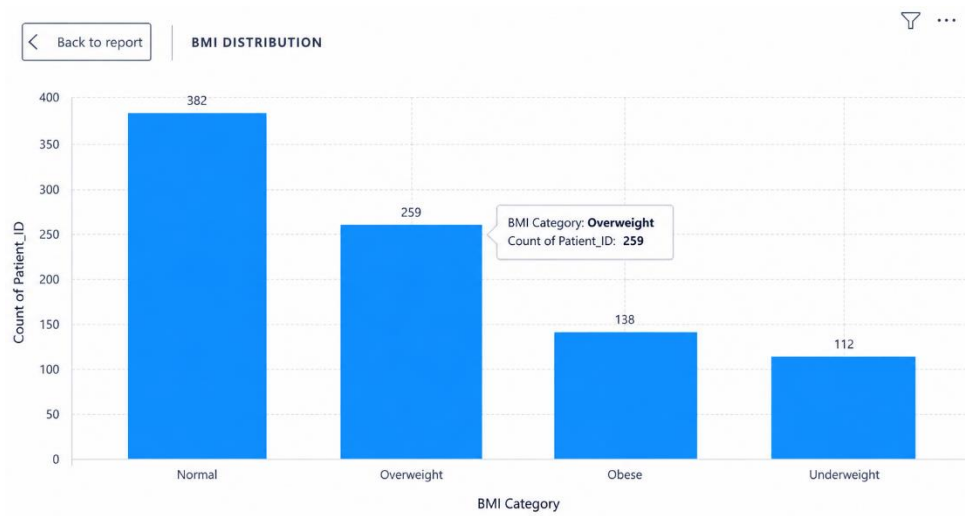
Age Group Distribution

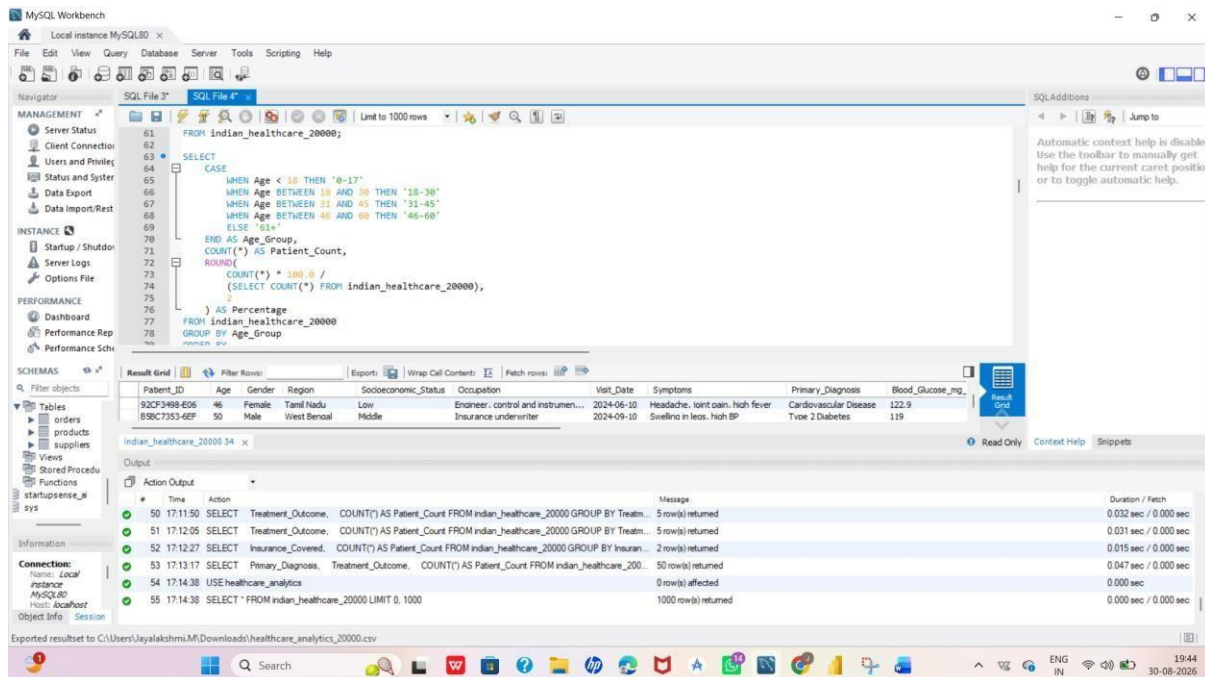
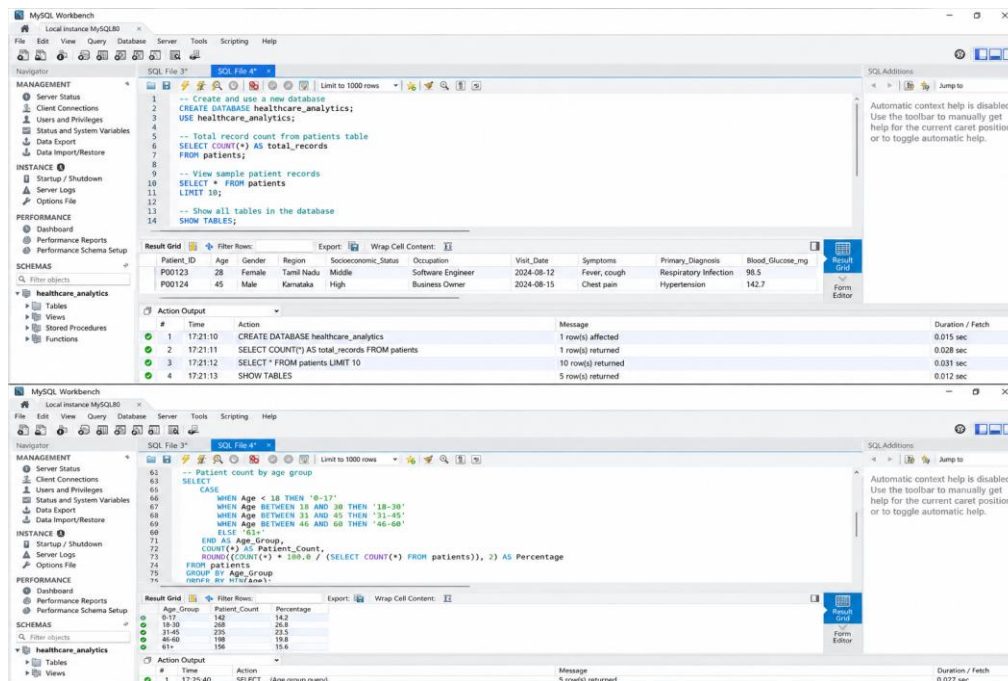


Region-wise Patient Distribution



BMI Distribution





RESULT:

Thus, the Indian healthcare data was successfully analyzed using **MySQL** and visualized using **Power BI**. The project helped to identify patterns in patient demographics, diseases, treatments, treatment outcomes, BMI, and insurance coverage. The interactive dashboard provides a simple and clear way to understand the healthcare data.